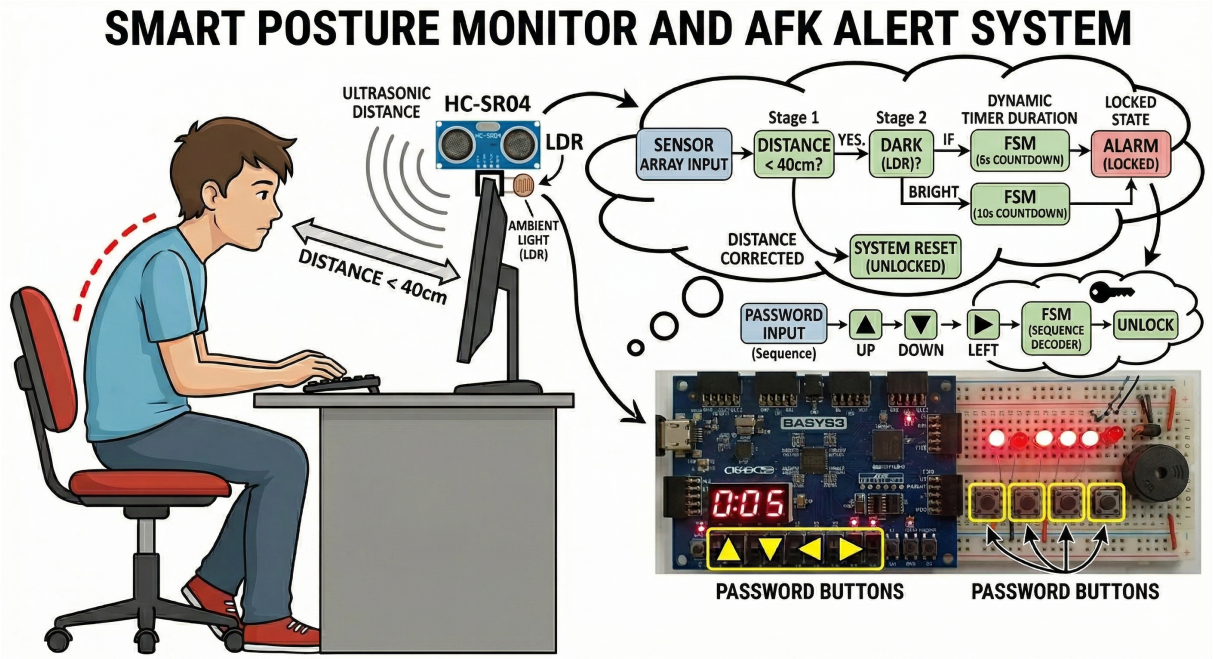
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Smart Posture Monitor and AFK Alert System

Abstract

The purpose of this term project is to design a smart, autonomous posture monitoring system using the Basys3 FPGA board, an ultrasonic sensor, and a photoresistor (LDR). Often, individuals lose track of their posture and lean dangerously close to their monitors during intensive computer sessions. This project solves that problem by continuously measuring the distance between the user and the monitor. To increase the complexity and efficiency of the system, an environmental light detection feature is integrated. If the user stays below a certain distance threshold in a well-lit room, a 10-second timer starts. However, if the room is dark—which causes higher eye strain—the timer is dynamically reduced to 5 seconds. Furthermore, a sequence detector logic is implemented; once the alarm is triggered, the system locks itself and can only be dismissed by entering a specific sequence using the directional buttons on the Basys3 board, demonstrating advanced practical applications of digital logic, multiplexers, and finite state machines (FSM).



Design Specification Plan

There will be several hardware components used in this project. The Basys3 board will be serving as the central control unit and logic processor. The PMOD pins of the Basys3 board will be utilized to connect the external components. The primary component for the system is the HC-SR04 Ultrasonic Sensor, which enables the measurement of distance in real-time. An LDR (Photoresistor) combined with a voltage divider circuit will be used to detect the ambient light level (dark or bright). A breadboard will be used to make the proper connections and organize jumper cables safely. The 4-digit 7-Segment Display will demonstrate the real-time distance in centimeters or act as a countdown timer. Furthermore, the built-in LEDs, an external buzzer, and the directional push buttons on the Basys3 board will be used as warning mechanisms and password inputs.

Methodology

The system will rely heavily on an internal Finite State Machine (FSM) implemented in VHDL. A custom clock_divider will be used to create accurate 1-second ticks from the Basys3's 100MHz clock. The ultrasonic sensor will continuously calculate the distance based on the echo pulse width. Simultaneously, the LDR will provide a logical '1' or '0' depending on the room's brightness.

When the calculated distance falls below the safe threshold (e.g., 40 cm), the FSM will check the LDR status. If it is bright, a 10-second countdown will begin; if it is dark, a 5-second countdown will begin. If the user corrects their posture before the timer expires, the system resets. However, if the limit is exceeded, the FSM transitions to a "Locked Alarm State," activating the LEDs and the buzzer. To exit this state and silence the alarm, the user must input a specific password sequence (e.g., Up -> Down -> Left -> Right) using the directional push buttons on the Basys3 board. This acts as a hardware sequence detector.

Function of BASYS3

The Basys3 board is the core of this project. The 4-digit 7-segment display will be driven dynamically by a multiplexer logic to show the real-time distance. The built-in LEDs will act as a visual alert system, flashing sequentially when the posture time limit is exceeded. The directional push buttons will function as the input interface for the sequence detector to unlock the alarm. Most importantly, the internal FPGA chip will process the entire logic—including the FSM, sequence detection, clock division, and distance calculations—strictly via VHDL, without relying on any external microcontrollers.

Project Demos

Progress Demo: For the progress demo, the ultrasonic sensor and the LDR modules will be fully functional. It will be ensured that the Basys3 board can successfully trigger the HC-SR04 sensor, calculate the distance in centimeters, and display this value dynamically on the 4-digit 7-segment display. Additionally, the system will successfully detect whether the

environment is dark or bright using the LDR, and this status will be represented by one of the built-in LEDs.

Final Demo: For the final demo, the complete FSM, the dynamic countdown logic (10s/5s), the external buzzer, and the sequence detector logic will be implemented. The final hardware connections on the breadboard will be established. It will be demonstrated that the system transitions to the locked alarm state correctly when the specific posture and light thresholds are violated. Furthermore, the alarm will only be deactivated when the correct button sequence is entered. Overall, the full system will be tested and debugged to ensure the complex autonomous design works properly.

Components

- Basys3 FPGA Board
- HC-SR04 Ultrasonic Proximity Sensor
- LDR (Photoresistor)
- 10k Ohm Resistor (for LDR voltage divider)
- 5V Active Buzzer
- Breadboard
- Jumper Cables (Male-to-Female & Male-to-Male)